

VNUHCM - University of Science
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CS486 – Lecture 4

Query Languages

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Content

- Introduction
- Define relation schemas
- Manipulate on relation schemas
 - Data modification
 - Data retrieval
 - Basic query
 - * **On one schema**
 - * On two or many schemas
 - Advanced query
- Summary

In relational algebra

- **Selection**
- **Projection**
- Cartesian product
- Join operation
- Set operations
- Division operation
- Other operations
- Modification operations

Selection

- Is applied to relation R to produce a new relation with a subset of R 's tuples
- Tuples of resulting relations satisfy some condition C
- Denotation $\sigma_c(R)$
- C is a Boolean expression made up of clauses
 - <attribute> <comparison operator> <constant>
 - <attribute> <comparison operator> <attribute>
 - <comparison op> : < , > , ≤ , ≥ , ≠ , =
 - Clauses are connected by Boolean operator : ∧ , ∨ , ¬

Selection

- The result is a relation
 - The same list of attributes as R
 - The number of tuples is less than or equal to the number of tuples of R

- Example

R	A	B	C	D
	α	α	1	7
	α	β	5	7
	β	β	12	3
	β	β	23	10

$$\sigma_{(A=B) \wedge (D > 5)}(R)$$

A	B	C	D
α	α	1	7
β	β	23	10

Selection

- Selection operator is commutative

$$\sigma_{c_1}(\sigma_{c_2}(R)) = \sigma_{c_2}(\sigma_{c_1}(R)) = \sigma_{c_1 \wedge c_2}(R)$$

Example 1

- List all employees who work in the department 4
 - Relation: EMPLOYEE
 - Attribute: DNo
 - Condition: DNo=4

Example 2

- Select tuples for all employees who either work in department 4 and make over \$25,000 per year or work in department 5 and make over \$30,000
 - Relation: EMPLOYEE
 - Attributes: Salary, DNo
 - Condition:
 - (Salary>25000 and DNo=4) or
 - (Salary>30000 and DNo=5)

Projection

- Is used to produce from a relation R a new relation that has only some of R 's columns
- Denotation $\pi_{A_1, A_2, \dots, A_k}(R)$
- The result is a relation
 - Has k attributes
 - The number of tuples is less than or equal to the number of tuples of R

- Example

R	A	B	C
	α	10	1
	α	20	1
	β	30	1
	β	40	2

$\pi_{A,C}(R)$

A	C
α	1
α	1
β	1
β	2

Projection

- Projection operator is not commutative

$$\pi_{X,Y}(R) \neq \pi_X(\pi_Y(R))$$

$$\pi_{A1, A2, \dots, An}(\pi_{A1, A2, \dots, Am}(R)) = \pi_{A1, A2, \dots, An}(R), n \leq m$$

Example 3

- List out the name and salary of employees

Extended projection

- Extending the projection operator to allow it to compute with components of tuples
- Denotation $\pi_{F_1, F_2, \dots, F_n}(E)$
 - E is a relation algebra expression
 - $F_1, F_2, \dots, F_n$ are arithmetic/string expressions involving
 - Attributes in E
 - Constants
 - Arithmetic operators ($a + b$: sum, a/b , $a*b$, $a - b$)
 - String operators ($c \parallel d$: concatenate)

Example 4

- List out the employees' name and salary after increasing by 10%

Sequences of operations

- Apply several relational algebra operations one after one
 - A single relational algebra expression

$$\pi_{A1, A2, \dots, Ak}(\sigma_C(R))$$

$$\sigma_C(\pi_{A1, A2, \dots, Ak}(R))$$

- Break down a complex expression into simpler steps
 - Step 1 $\sigma_C(R)$
 - Step 2 $\pi_{A1, A2, \dots, Ak}(\text{the result of step 1})$

↓
Giving a name

Assignment operator

- Is often used to receive the result of an operation
 - The intermediate result in a sequence of operations
- Denotation $\leftarrow$

- Example

- Step 1: $S \leftarrow \sigma_c(R)$

- Step 2:

RESULT $\leftarrow \pi_{A_1, A_2, \dots, A_k}(S)$

Rename operator

- Rename either the relation name or attribute name

- Relation

Examine $R(B, C, D)$

$\rho_S(\mathbf{R})$: Rename the name of relation R to S

- Attribute

$\rho_{X, C, D}(\mathbf{R})$: Rename the name of attribute B to X

Rename the name of relation R to S and the name of attribute B to X

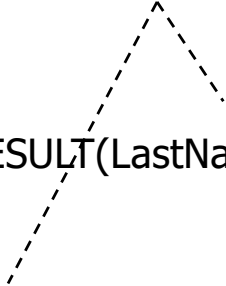
$\rho_{S(X, C, D)}(\mathbf{R})$

Example 5

- List out the name of employees who work in the department 4

- Case 1: $\pi_{\text{LName, Minit, FName}}(\sigma_{\text{DNo}=4}(\text{EMPLOYEE}))$

- Case 2: $\text{EMP_DEP4} \leftarrow \sigma_{\text{DNo}=4}(\text{EMPLOYEE})$
 $\text{RESULT} \leftarrow \pi_{\text{LName, Minit, FName}}(\text{EMP_DEP4})$


$$\rho_{\text{RESULT}}(\text{LastName, MidName, FirstName})(\pi_{\text{LName, Minit, FName}}(\text{EMP_DEP4}))$$

$$\text{RESULT}(\text{LastName, MidName, FirstName}) \leftarrow \pi_{\text{LName, Minit, FName}}(\text{EMP_DEP4})$$

In SQL

- Basic query is formed of the three clauses

```
SELECT <List_of_columns>  
FROM   <List_of_tables>  
[WHERE] <Condition>
```

- <List_of_columns>
 - Column names showed in the result of the query
- <List_of_tables>
 - Table names required to process the query
- <Condition>
 - Boolean expression that identifies the rows to be retrieved
 - Expression's connection : AND, OR, and NOT
 - Operations: < , > , <=, >=, <>, =, LIKE and BETWEEN

SQL and Relational Algebra

~~SELECT~~ <List_of_columns>
~~FROM~~ <List_of_tables>
~~WHERE~~ <Condition>

SELECT L
FROM (σ_R(R)) →
WHERE C

Example 1

- List all employees who work in the department 4

SELECT clause

- List the name of male employees who work in the department 4

SELECT clause

Alias name

```
SELECT LName AS 'Last Name', Minit AS 'Middle Name',  
       FName AS 'First Name'  
FROM   EMPLOYEE  
WHERE  DNo=4 AND Sex= 'Male'
```

$$R1 \leftarrow (\pi_{LName, Minit, FName}(\sigma_{DNo=4 \wedge Sex='Male'}(EMPLOYEE)))$$
$$\rho_{RESULT(Last\ Name, Middle\ Name, First\ Name)}(R1)$$

SELECT clause

Extension

```
SELECT LName + ' ' + Minit + ' ' + FName AS 'Full Name'  
FROM   EMPLOYEE  
WHERE  DNo=4 AND Sex= 'Male'
```

$$\rho_{\text{Full Name}}(\pi_{\text{LName}||\text{Minit}||\text{FName}}(\sigma_{\text{DNo}=4 \wedge \text{Sex}='Male'}(\text{EMPLOYEE})))$$
$$\rho_{\text{SSN}, \text{Full Name}}(\pi_{\text{LName}+\text{Minit}+\text{FName}}(\sigma_{\text{DNo}=4 \wedge \text{Sex}='Male'}(\text{EMPLOYEE})))$$


SELECT clause

Extension

```
SELECT SSN, Salary*1.1 AS '10%SalaryIncrease'  
FROM EMPLOYEE  
WHERE DNo=4 AND Sex= 'Male'
```

$$\rho_{SSN, 10\%SalaryIncrease}(\pi_{SSN, Salary*1.1}(\sigma_{DNo=4 \wedge Sex='Male'}(EMPLOYEE)))$$

SELECT clause

Duplicate tuples are eliminated

```
SELECT DISTINCT Salary
FROM EMPLOYEE
WHERE DNo=4 AND Sex= 'Male'
```

Salary
30000
25000
28000
38000

- Cost
- Users want to see all tuples

WHERE clause

BETWEEN

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE Salary >= 20000 AND Salary < 30000
```

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE Salary BETWEEN 20000 AND 30000
```

WHERE clause

NOT BETWEEN

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE Salary NOT BETWEEN 20000 AND 30000
```

WHERE clause

LIKE

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE Address LIKE 'Frondren _ _ _ _'
```

Arbitrary characters

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE Address LIKE 'Frondren %'
```

Arbitrary strings

WHERE clause

NOT LIKE

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE LName LIKE 'Smith'
```

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE LName NOT LIKE 'Smith'
```

WHERE clause

ESCAPE

```
SELECT SSN, FName  
FROM   EMPLOYEE  
WHERE  Address LIKE '123x/%Frondren' ESCAPE 'x'
```

WHERE clause

Datetime

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE BDate BETWEEN '1955-12-08' AND '1966-07-19'
```

'1955-12-08'	YYYY-MM-DD	' 17:30:00'	HH:MI:SS
' 12/08/1955'	MM/DD/YYYY	' 05:30 PM'	
'December 8, 1955'			

'1955-12-08 17:30:00'

WHERE clause

NULL

- SQL allows attributes to have value NULL
 - Value unknown (eg. birthday)
 - Value inapplicable (eg. name of spouse)
 - Value withheld (eg. phone number)
- Operation on a NULL and any value, the result is NULL
 - x has a value NULL
 - $x + 3$ is also NULL
 - $x + 3$ is not a legal SQL expression
- Comparison on a NULL value and any value, the result is UNKNOWN
 - The value of $x = 3$ is UNKNOWN
 - The comparison $x = 3$ is not correct SQL

WHERE clause

NULL

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE SuperSSN IS NULL
```

```
SELECT SSN, FName  
FROM EMPLOYEE  
WHERE SuperSSN IS NOT NULL
```

WHERE clause

UNKNOWN

- The comparison involving NULL will result the three-value logic
 - True (1)
 - False (0)
 - Unknown (1/2)
- Logical operator
 - x and y (Minimum value)
 - x or y (Maximum value)
 - not x (1-x)
- The condition in WHERE clauses will result **False** if tuples with **Unknown** value

In relational calculus

- Tuple relational calculus (TRC)
- Domain relational calculus (DRC)

Tuple relational calculus

- A simple tuple calculus query is of the form

$$\{ t.A \mid P(t) \}$$

- t is a tuple variable
 - Its value is any individual tuple from a relation
 - $t.A$ is a value of a tuple t at an attribute A
- P is a conditional expression involving t
 - $P(t)$ has the TRUE or FALSE value depending on t
- The result
 - The set of all tuples t that satisfy $P(t)$

Example 1

- Find employees who work in the department 4

$$\{ t \mid \underbrace{t \in \text{EMPLOYEE}}_{P(t)} \wedge \underbrace{t.\text{DNO} = 4}_{P(t)} \}$$

- $t \in \text{EMPLOYEE} : \text{TRUE}$
 - If t is an instance of relation EMPLOYEE
- $t.\text{DNO} = 4 : \text{TRUE}$
 - If the attribute DNO of tuple t has a value being equal to 4

Example 1'

- Retrieve the SSN and the first name of employees who work in the department 4

Domain relational calculus

- An expression of the domain calculus is of the form

$$\{ x_1, x_2, \dots, x_n \mid P(x_1, x_2, \dots, x_n) \}$$

- $x_1, x_2, \dots, x_n$ are domain variables
 - Accepting single values from the domain of attributes
- P is a formula of variables $x_1, x_2, \dots, x_n$
 - P is formed from atoms
- The result
 - The set of values such that when assigned to variables x_i , they make P TRUE

Example 1'

- Retrieve the SSN and the first name of employees who work in the department 4

Next lecture

- Data retrieval
 - Basic query
 - **On two or many schemas:** Cross product & join operation
- Read
 - Chapter 5 [The complete book, 2E]
 - Chapter 6, 8 [Fundamentals of Database System, 7E]
- Watch
 - Video clip

